separately introduced inside the engine combustion chamber and then ignited. Hence for supplying the required amount of fuel (in diesel engine) or fuel air mixture (in petrol engines) depending upon the engine's speed and load governor unit is used. At all speeds and loads to operate the engine safely the governor has the following operation stages of operation,

1. Governor cut in speed
2. Governor cut out speed
3. Governor over run

Note:

Governor Cut in Speed: Governor cut in speed represents controlling the speed of the engine by using the control rack by moving the rack out and supplying the required amount of fuel to the engine.

Governor cut out speed: When the control rod is operated beyond the control speed the fuel supply is cut by the governor this is called as cut out speed.

Governor Over run: The difference between the governor cut in and cut out speeds is called as governor over run.

9.10.1 Governor Terminologies

Based on the working principle governors can be classified into three types. They are

1. Mechanical Governors
2. Pneumatic Governors
3. Hydraulic Governors

1. **Mechanical Governor:** To control the engine's speed and load and reduce wastage of fuel governor is used in engines. The mechanical governor is operated mechanically. This governor is generally used in diesel engines. The construction and working principle of the mechanical governor are as follows,

   Construction: In this governor spring loaded weights, control rack, sleeve and bell crank lever are present and linked together properly. The sleeve is placed on the governor shaft and allowed to move freely. The two centrifugal masses are connected with the plunger sleeve through the bell crank lever. The governor shaft gets the power from the engine. The layout of the mechanical governor can be seen in Fig 9.10.

1. ***Height of a governor.*** *It is the vertical* distance from the centre of the ball to a point where the axes of the arms (or arms produced) intersect on the spindle axis. It is usually denoted by *h*.
2. ***Equilibrium speed.*** *It is the speed at* which the governor balls, arms etc., are in complete equilibrium and the sleeve does not tend to move upwards or downwards.
3. ***Mean equilibrium speed.*** *It is the speed* at the mean position of the balls or the sleeve.
4. ***Maximum and minimum equilibrium speeds.*** *The speeds at the maximum and minimum* radius of rotation of the balls, without tending to move either way are known as maximum and minimum equilibrium speeds respectively.
5. ***Sleeve lift.*** *It is the vertical distance which* the sleeve travels due to change in equilibrium speed.

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